

From Heuristics to Geometry: Re-engineering Platform Profitability through Dynamic Optimization

1. The Economic Imperative: From Structural Loss to Positive Margins

In the competitive landscape of ride-pooling, the transition from static, rule-based heuristics to dynamic geometric optimization is a strategic necessity. Historically, our platform relied on "local convenience" matching—prioritizing immediate dispatch and proximity. While operationally simpler, this approach created a "structural loss" of -\$0.141 per request. The only viable path to the current \$0.135 profit per request is a shift toward a density-focused geometric model.

This evolution is defined by a \$3,266.33 total profit swing, achieved by moving away from indiscriminate volume in favor of high-quality, selective matching.

Metric	Historical Baseline	Optimized Policy (120s Wait)	Delta Analysis	Directional Impact
Total Profit	-\$1,669.75	+\$1,596.58	+\$3,266.33	Turnaround
Profit per Request	-\$0.141	+\$0.135	+\$0.277	Margin Shift
Shared-Rate Given Conversion	79.8%	68.5%	-11.3%	Selective Quality
Avg. Wait Time (s)	26.8s	51.5s	+24.7s	Strategic Investment

The increase in average wait time (from 26.8s to 51.5s) is not a service failure, but a strategic investment in demand density. By extending the temporal window, we accumulate the spatial density required for the matching engine to identify truly efficient routes. To enable this engine, however, we must first implement pricing mechanisms that prepare the market and protect the input stream of riders.

2. Strategic Pricing: Geometric Weighting and Conversion Protection

Pricing serves as both a "volume builder" and a "margin protector" within our spatial density framework. It is the lever used to incentivize riders into the queue while ensuring the platform is not exposed to unprofitable dispatches.

Baseline Calibration

We established a base price of 0.725, calibrated specifically against the platform's fixed driving cost of \$0.70 per mile. This 0.025 buffer ensures that even if a rider is dispatched solo (the worst-case scenario for a pooling engine), the platform maintains a positive margin. Prices are adjusted around this baseline—discounting short trips to build density and pricing long trips to safeguard against complex routing costs.

Pricing Constraints and Rationale

To maintain a stable candidate pool for the matching engine, we clip prices within a strictly defined safe range to avoid demand evaporation.

Constraint	Value	Rationale
Lower Bound	0.62	Floor that prevents losses even during high-efficiency shared rides.
Upper Bound	0.81	Clipping point to stay safely below the "Conversion Cliff."
Global Max Accepted Quote	0.845	Historical ceiling where rider acceptance probability nears zero.

The Conversion Cliff and Protective Clipping

Our "Shadow Evaluator" identified a sharp "Conversion Cliff" governed by an **Exponential Decay Factor (-18.0)**. This factor models the extreme price elasticity of riders; once quotes exceed the historical "Support Cap," conversion probability collapses exponentially. By clipping prices at 0.81, we protect the input stream. If pricing is too aggressive, the queue thins, density drops, and the risk-pooling benefits evaporate.

3. The Selective Matching Engine: A Two-Stage Hierarchical Approach

The shift from "local convenience" to "intentionally selective" matching requires a two-stage hierarchy to balance computational efficiency with geometric precision.

Stage 1: Fast Candidate Shortlisting

To manage real-time computational overhead, the engine avoids performing expensive exact routing calculations for every possible pair. Instead, it uses a "cheap" compatibility score to filter the queue, identifying the **top 12 candidates** by geographic and temporal similarity. This shortlist provides a statistically sufficient pool for Stage 2.

The compatibility score:

$$q(i,j) = 2|a^p_i - a^p_j| + 2|a^d_i - a^d_j| + 0.003|t_i - t_j| + 0.2|\ell_i - \ell_j|$$

Notation	Meaning
a^p_i	pickup area of rider i
a^d_i	dropoff area of rider i
t_i	arrival time of rider i
ℓ_i	solo trip length of rider i

Stage 2: Exact Match Scoring $M(i, j)$

The shortlisted candidates are then evaluated through a weighted scoring function designed to prioritize marginal profit:

$$M(i,j) = 2.2 \cdot \text{SaveRatio}_{ij} + 0.35 \cdot \text{SharedRatio}_{ij} + 0.18 \cdot \text{AbsSave}_{ij} - 0.18 \cdot \text{DetourPenalty}_{ij}$$

- **SaveRatio (Weight 2.2):** This is the primary driver of Vehicle Miles Traveled (VMT) reduction. A weight of 2.2 acts as a "geometric threshold," ensuring that the engine only executes matches where the distance saved is significant enough to outweigh operational costs.
- **Detour Penalty (Weight -0.18):** This coefficient penalizes matches that require inefficient solo segments. At \$0.70/mile, detours are costly; this penalty ensures "locally convenient" matches are rejected if they erode the platform's bottom line.

This hierarchy ensures that we do not sacrifice "SaveRatio" priority for the sake of speed, though its efficacy remains entirely dependent on the batching horizon.

Dynamic Matching Thresholds

We also use minimum quality thresholds to avoid matching a pair only because it has the highest score. The pair must also pass minimum quality thresholds:

$$\text{SaveRatio}_{ij} \geq \tau_r$$

$$\text{AbsSave}_{ij} \geq \tau_a$$

By default, $\tau_r = 0.12$ and $\tau_a = 0.35$. In other words, the pair must save at least 12% of the distance and at least 0.35 miles.

However, these thresholds are relaxed when matching becomes more urgent:

- When the queue is large, $|s| \geq 8$, many riders are waiting, so delaying too much may lead to costly solo dispatches.
- When the rider arrives late in the time window, $t_i > 3300$, there is less time to wait for a better future match.

Even after relaxation, the policy keeps hard lower bounds:

$$\tau_r \geq 0.05$$

$$\tau_a \geq 0.12$$

This prevents the policy from accepting very weak matches.

Final Matching Decision

Among candidates that pass both thresholds, the policy selects the one with the highest exact score:

$$j^* = \operatorname{argmax}_{j \in C_i(s)} M(i, j)$$

subject to:

$$\text{SaveRatio}_{ij} \geq \tau_r$$

$$\text{AbsSave}_{ij} \geq \tau_a$$

If such a candidate exists, the incoming rider is matched with j^* . If no candidate satisfies the thresholds, the policy returns None, meaning the rider waits.

4. Maximizing the Power of Pooling: Batching Horizons and Demand Density

The "Temporal Sweep" analysis across a "Wait Grid" (15s, 60s, 120s) identifies the batching horizon as the primary lever for capturing spatial density.

Comparing the 15-second and 120-second horizons reveals the "Power of Pooling": at 15 seconds, the lack of density results in a **Shared-Rate Given Conversion of only 28.3%**. Extending the horizon to 120 seconds allows the rate to climb to **68.5%**.

Crucially, the optimized policy's 68.5% shared-rate is actually lower than the historical baseline of 79.8%. This represents a **deliberate shift toward selectivity**. While the historical model accepted nearly 80% of matches regardless of geometric efficiency (leading to structural loss), our model rejects low-

quality pairings. The 120-second horizon is the "Sweet Spot": the marginal gain in demand density—and thus the ability to find high-SaveRatio matches—outweighs the marginal risk of rider attrition. This strategy was validated through rigorous simulation.

5. Validation Architecture: The Shadow Evaluator and Conversion Proxies

To bridge the gap between theory and reality, we utilized the "Shadow Evaluator," a discrete-event simulation environment that replayed **11,788 historical arrivals**.

The simulation uses an **Empirical Conversion Proxy** consisting of 20 Price Buckets (in \$0.05 increments). These buckets allow the model to aggregate historical rider behavior into stable probability tables, ensuring sensitivity to price shifts while maintaining statistical robustness.

Realism is enforced through the "**Support Cap**" logic. By applying the **Exponential Decay Factor (-18.0)** to any quote exceeding historically accepted rates, the simulation penalizes unrealistic pricing. This architecture serves as the empirical test for our stochastic inventory theories, providing the data-driven confidence necessary for deployment.

6. Theoretical Foundation: Risk Pooling and Stochastic Optimization

Our results are grounded in the "Risk-Pooling Effect" (Eppen 1979). In a decentralized, heuristic system, every dispatch acts as an independent newsvendor. By centralizing and batching demand through a 120-second window, we reduce the coefficient of variation in demand. Because safety stock (excess VMT) is a **concave function of demand variance**, this centralization allows the platform to meet its service levels with significantly less "inventory" in the form of empty vehicle miles.

Applying **Newsvendor Logic** to ride-pooling, we define the costs as follows:

- **Overage (c_o):** The cost of excess capacity, quantified as the \$0.70/mile driving cost for empty or inefficient miles.
- **Underage (c_u):** The opportunity cost of lost revenue and platform utility when a rider rejects a quote.

By prioritizing the "SaveRatio," we align the matching engine with the maximization of the objective function $T(y)$, ensuring concavity and an optimal profit solution. Furthermore, the **Envelope Theorem** confirms that our profit shift is a direct consequence of tuning the batching horizon and pricing parameters. It allows us to understand how the platform's value function (Profit) responds to these parameters without needing to re-solve the entire matching optimization at every step.

7. Strategic Conclusion: The Future of Platform Profitability

The fundamental takeaway of this re-engineering effort is that ride-pooling profitability is maximized through **geometric alignment** rather than volume alone. By prioritizing quality over instantaneity, we have fundamentally altered the platform's economic trajectory.

The three most critical strategic pillars are:

1. **Geometric Weighting vs. Rules:** The "SaveRatio" is the vital metric for reducing VMT and driving margin.
2. **Density over Instantaneity:** The 120s batching window is a mandatory strategic investment for capturing poolable density.
3. **Simulation-Driven Confidence:** Stress-testing against conversion proxies and the "Conversion Cliff" ensures operational reality.

Through these advancements, the platform has successfully transitioned from a loss-making historical baseline to a robust, \$0.135 profit-per-request model.